

6. Finite Word length Effects

6.2 6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB

OUTPUT:

#Code

```
clc;
```

```
clear;
```

```
close all;
```

```
% Parameters
```

```
A = 1;           % Amplitude
```

```
f = 5;           % Frequency (Hz)
```

```
Fs = 100;        % Sampling frequency
```

```
n = 0:1/Fs:1;    % Time vector
```

```
% Generate sinusoidal signal
```

```
x = A*sin(2*pi*f*n);
```

```
% Number of quantization bits
```

```
bits = 4;
```

```
L = 2^bits;       % Number of quantization levels
```

```
% Uniform Quantization
```

```
xmin = -1;
```

```
xmax = 1;
```

```
delta = (xmax - xmin)/(L-1);
```

```
xq = round((x - xmin)/delta)*delta + xmin;
```

```
% Quantization Error
```

```
e = x - xq;
```

```
% Plotting
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(n,x,'b','LineWidth',1.5);
```

```
title('Original Sinusoidal Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
subplot(3,1,2);
```

```
stairs(n,xq,'r','LineWidth',1.5);
```

```
title('Uniformly Quantized Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
subplot(3,1,3);
```

```
plot(n,e,'k','LineWidth',1.5);
```

```
title('Quantization Error');
```

```
xlabel('Time (s)');
```

```
ylabel('Error');
```

```
grid on;
```

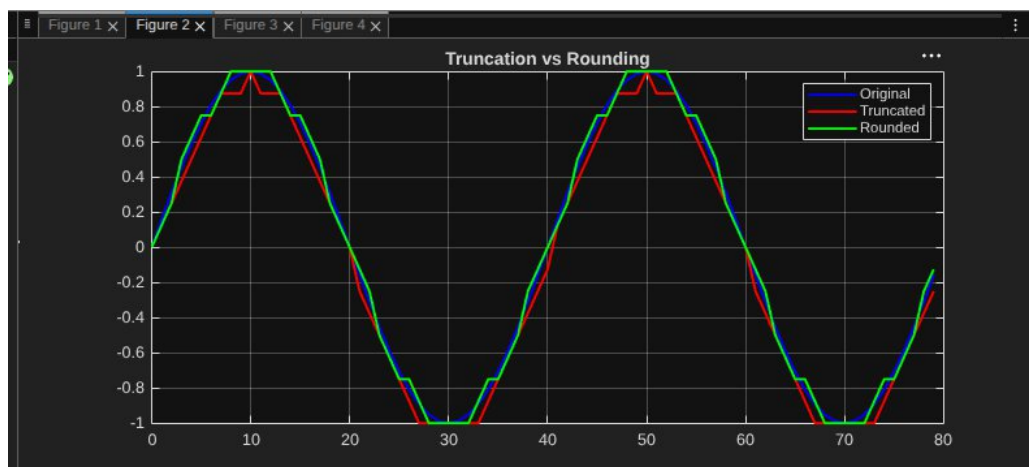
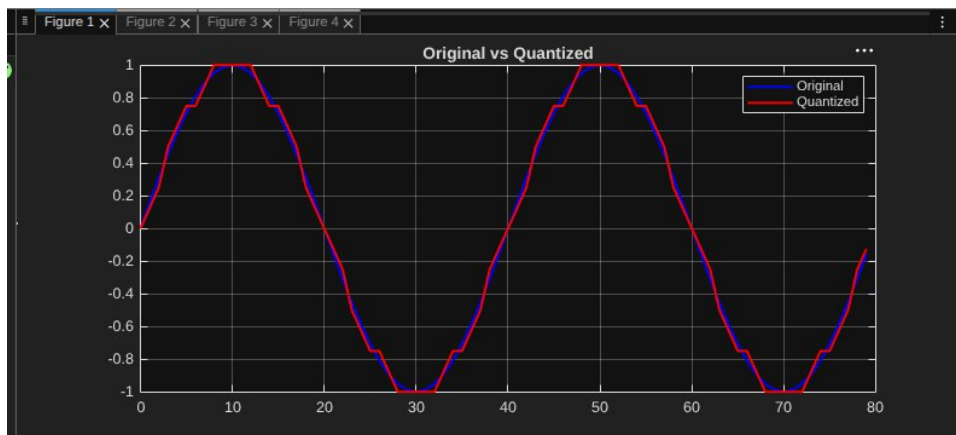
```
% Mean Square Error
```

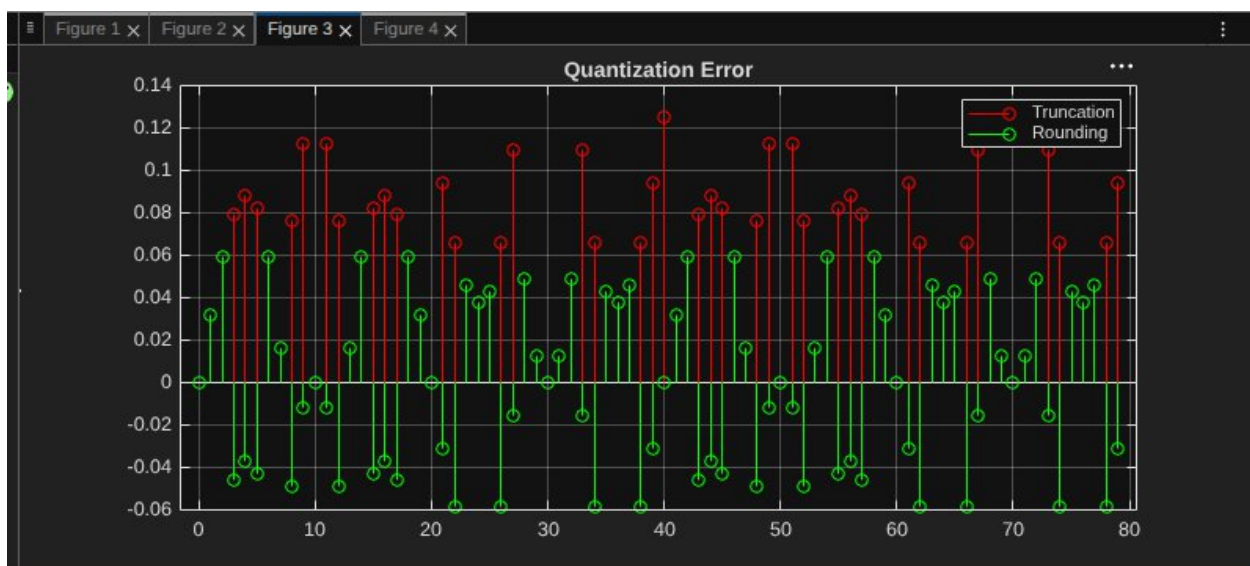
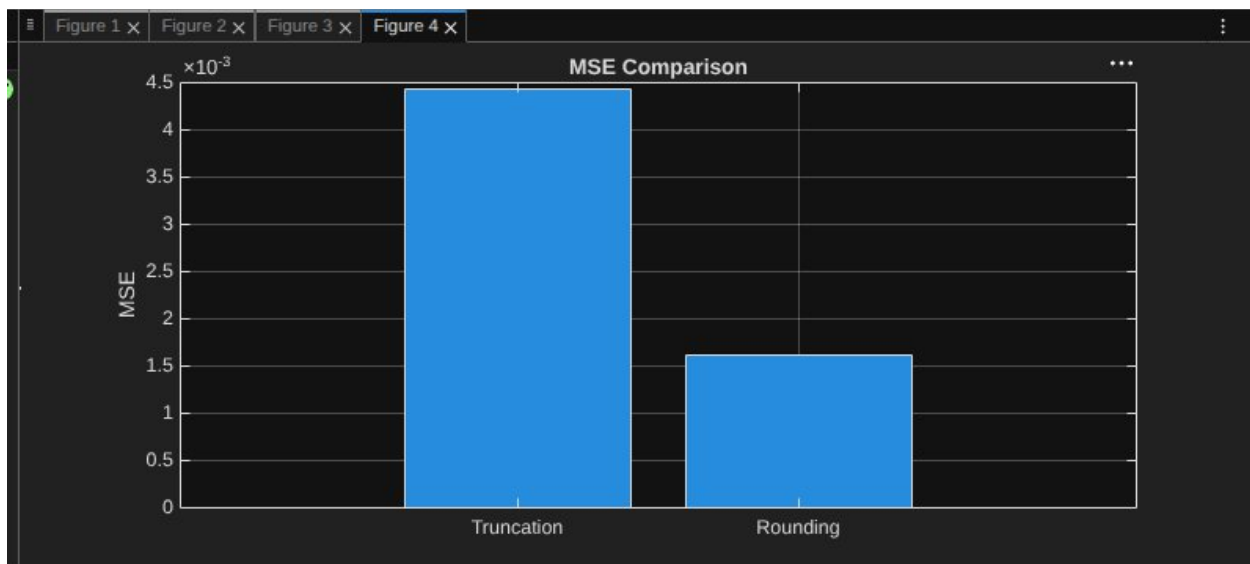
```
MSE = mean(e.^2);
```

```
fprintf('Quantization Step Size = %.4f\n',delta);
```

```
fprintf('Mean Square Error = %.6f\n',MSE);
```

#Output





6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB\

#Code:

```
clc;

clear;

close all;

%% Signal

fs = 1000;

f = 50;

t = 0:1/fs:0.1;

x = sin(2*pi*f*t);

%% Quantization Levels

L = [8 16 32];

colors = {'r','g','m'};

figure;

for i = 1:length(L)

    % Quantization

    xq{i} = round(((x + 1) * (L(i) - 1) / 2)) * (2 / (L(i) - 1)) - 1;

    % Error & MSE

    e{i} = x - xq{i};

    mse(i) = mean(e{i}.^2);

    % Plot Quantized Signals

    subplot(4,1,i+1);

    stairs(t, xq{i}, colors{i}, 'LineWidth', 1.2);

    title(['Quantized Signal (' num2str(L(i)) ' Levels)']);

    grid on;

end

%% Original Signal
```

```

subplot(4,1,1);

plot(t, x, 'LineWidth', 1.5);

title('Original Signal');

grid on;

%% Display MSE

fprintf('MSE for 8 Levels = %f\n', mse(1));

fprintf('MSE for 16 Levels = %f\n', mse(2));

fprintf('MSE for 32 Levels = %f\n', mse(3));

%% Error Plots

figure;

for i = 1:3

    subplot(3,1,i);

    plot(t, e{i}, 'LineWidth', 1.2);

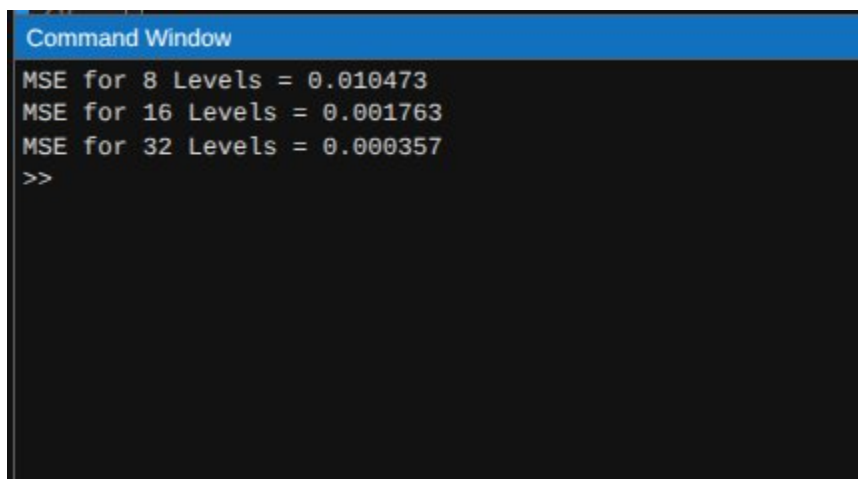
    title(['Error (' num2str(L(i)) ' Levels)']);

    grid on;

end

```

#Output



A screenshot of the MATLAB Command Window. The window has a blue title bar that says "Command Window". The background is black with white text. The text shows the results of the MATLAB code: "MSE for 8 Levels = 0.010473", "MSE for 16 Levels = 0.001763", and "MSE for 32 Levels = 0.000357". Below these lines, the prompt ">>" is visible, indicating the end of the command execution.

```

Command Window
MSE for 8 Levels = 0.010473
MSE for 16 Levels = 0.001763
MSE for 32 Levels = 0.000357
>>

```

